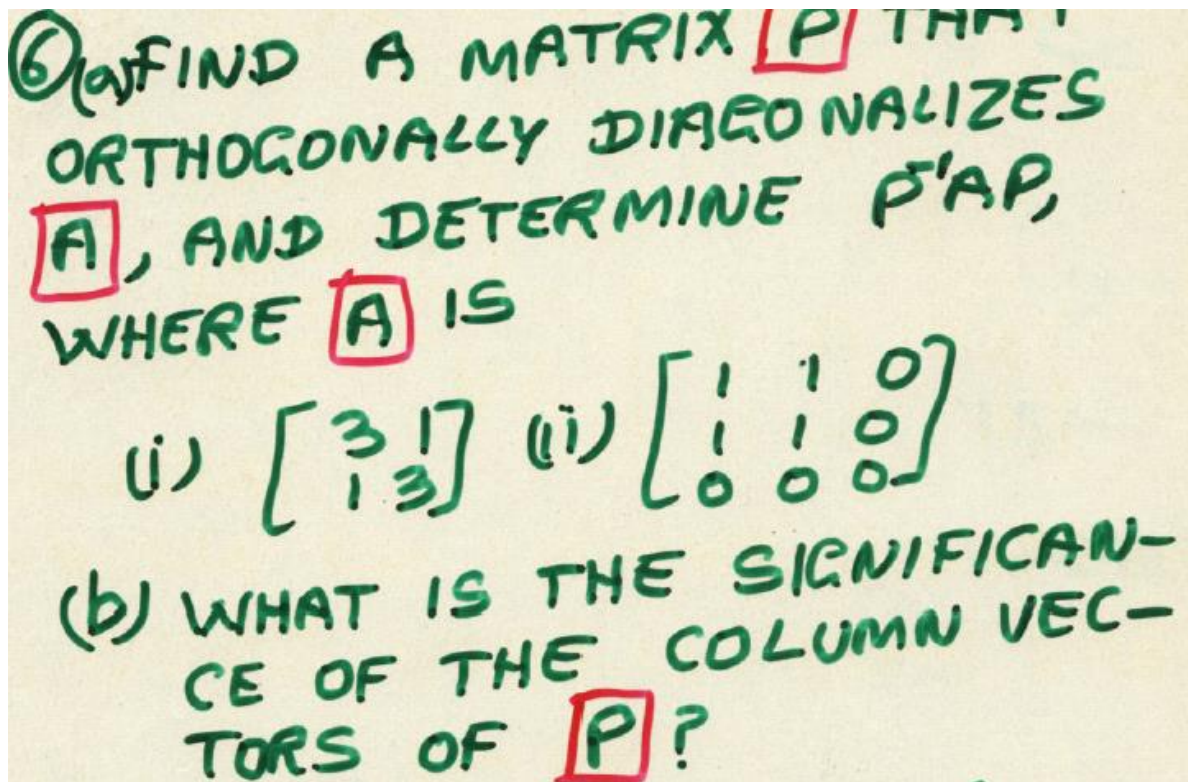


Assignment 6b(Q6,7,&8)

Thursday, 25 April 2024 1:10 am



Question 06

(a) $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

Solution: $\begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = (3-\lambda)^2 - 1 = 0 \Rightarrow \lambda^2 - 6\lambda + 8 = 0$

We have two eigenvalues $\lambda_1 = 4$ & $\lambda_2 = 2$. First take λ_1

$$\begin{bmatrix} 3-\lambda_1 & 1 & 0 \\ 1 & 3-\lambda_1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 3-4 & 1 & 0 \\ 1 & 3-4 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence we have $-x_1 + x_2 = 0$ and $x_2 = x_2$, so first eigenvector would be $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Similarly for λ_2 , we have second eigenvector $v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. For orthogonal, we need

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to each vector by its norm, so it would be unit vector...

$$v_1^* = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, v_2^* = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}. \text{ So, matrix } P \text{ would be } P = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$\text{and } P^{-1} = P^t = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$P^{-1} A P = P^t A P = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$$

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$$(b) \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Solution: Using similar procedure we have eigenvalues $\lambda_1 = 2, \lambda_2 = 0, \lambda_3 = 0$

Eigenvalue: 2, eigenvector: $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, so unit vector $\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$

Eigenvalue: 0, eigenvectors: $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, so they are orthogonal hence

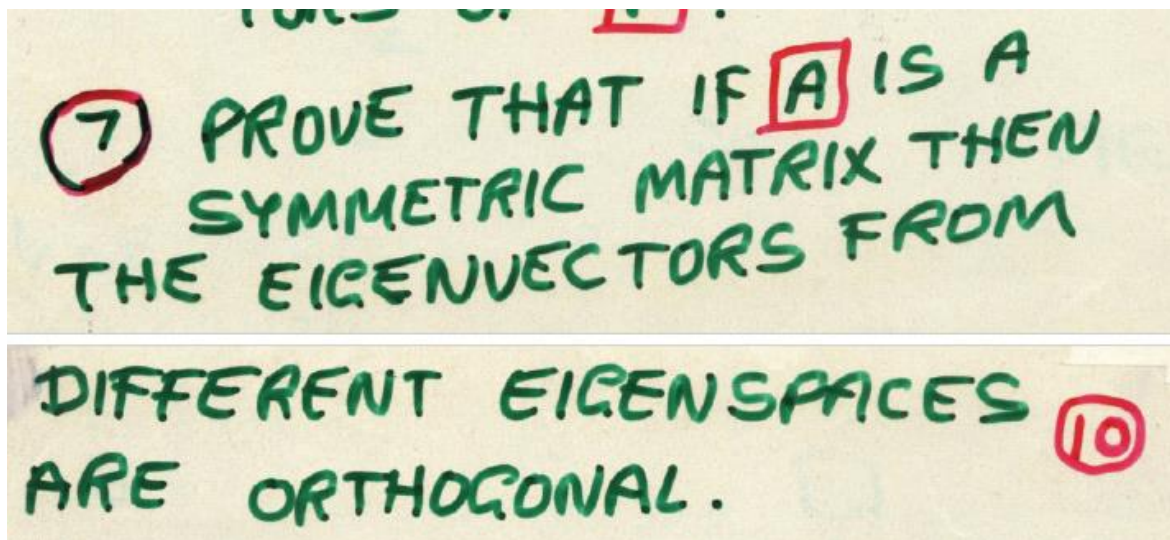
we only need to make them unit vector. Unit eigenvectors are $\begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$ and

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Form the matrix P , whose column i is eigenvector no. i : $P = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Form the diagonal matrix D whose element at row, column is eigenvalue no. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(b) Different Eigenvectors from different eigenvalues(Eigenspace) are orthogonal to each other, one can easily see in above examples.



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Question 07

If A is a real symmetric matrix, then any two eigenvectors corresponding to distinct eigenvalues are orthogonal.

Solution: Proof. Let λ_1 and λ_2 be distinct eigenvalues with associated eigenvectors v_1 and v_2 . Then, $Av_1 = \lambda_1 v_1$ and $Av_2 = \lambda_2 v_2$. Take the inner product of the first equation by v_2 and the inner product of the second equation by v_1 :

$$v_2^T A v_1 = \lambda_1 v_2^T v_1, \quad A v_2^T v_1 = \lambda_2 v_2^T v_1$$

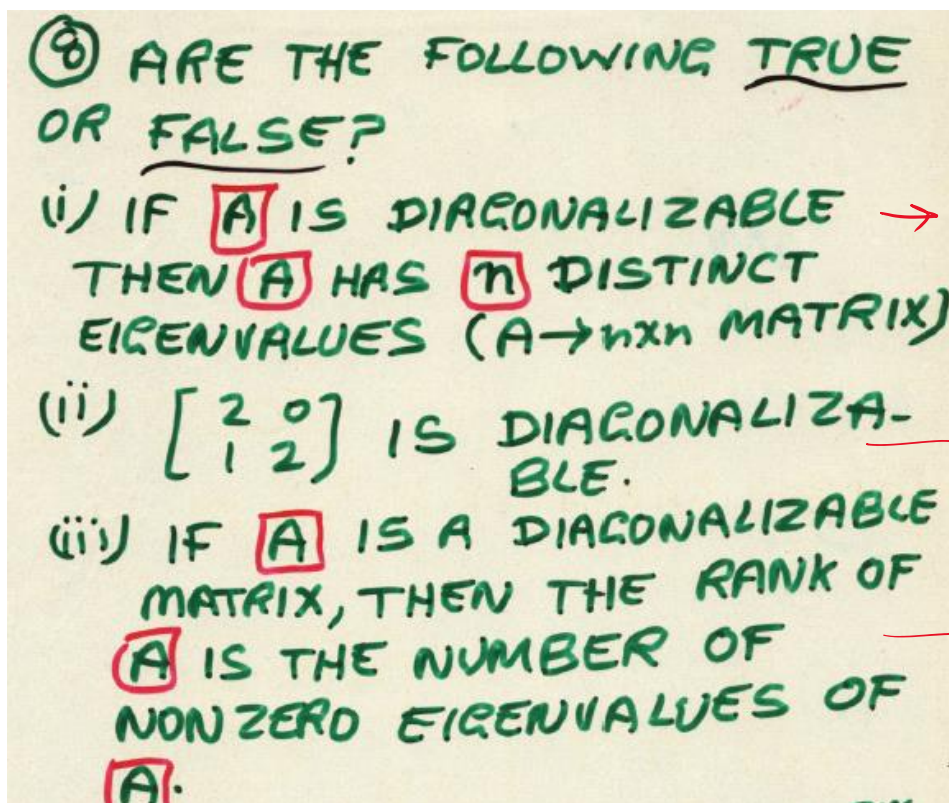
In Equation, $(A v_2)^T v_1 = v_2^T A^T v_1$, so becomes $v_2^T A v_1 = \lambda_1 v_2^T v_1$, $v_2^T A^T v_1 = \lambda_2 v_2^T v_1$. Since $A^T = A$, in Equation, we have $v_2^T A v_1 = \lambda_1 v_2^T v_1$, $v_2^T A v_1 = \lambda_2 v_2^T v_1$ and

$$\lambda_1 v_2^T v_1 = \lambda_2 v_2^T v_1$$

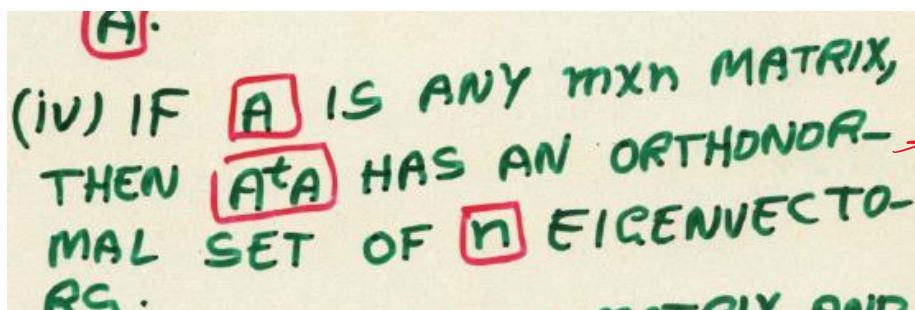
Equation gives

$$(\lambda_1 - \lambda_2) v_2^T v_1 = 0.$$

Since $\lambda_1 \neq \lambda_2$, $\langle v_2, v_1 \rangle = 0$, and v_1, v_2 are orthogonal.



$r(A) = r(P^{-1}AP) = r(D)$ = number of nonzero diagonal entries = number of non zero eigenvalues counting multiplicity,



MAL SET OF n EIGENVECTORS.
 (v) IF V IS ANY $n \times 1$ MATRIX AND
 I IS THE $n \times n$ IDENTITY MAT-
 RIX, THEN $I - VV^T$ IS ORTHO-
 GONALLY DIAGONALIZABLE.

$$(A^T A)^T \Rightarrow A^T (A^T)^T \Rightarrow A^T A$$

& It must have
 n -eigenvectors
 because it is diagonalizable

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$$(v) (I - VV^T)^T \Rightarrow I^T - (VV^T)^T$$

$\Rightarrow I - VV^T$ hence it is symmetric

So, If we have distinct Eigenvalues so the corresponding
 vectors would be orthogonally diagonalizable.